

pin is connected to GND, while for a common-anode RGB LED, it is connected to VCC. In this system, common-anode RGB LEDs have been used; therefore, the common terminal is connected to VCC. Table 3 explains the connections and functions of the various components used in the system.

Software

Two different codes are created: one for the controller and the other for the sensor. The sensor code continuously measures the distance

using an ultrasonic sensor in a loop and transmits the detected data via the I2C interface to another Arduino Nano. The controller device receives this data and controls the respective RGB LEDs based on traffic density and the predefined algorithm.

To implement this setup, the sensor unit is designated as the master Arduino and the receiving unit as the slave Arduino. Both codes are written and compiled using the Arduino IDE, which provides an easy interface for

writing, verifying, and uploading sketches. Before uploading, the correct board type (Arduino Nano) and COM port must be selected from the Tools menu in the Arduino IDE. The sensor code is then uploaded to the master Arduino and the controller code to the slave Arduino. Once programmed and connected through the I2C lines, the system automatically begins exchanging data. Fig. 4 shows the code snippet for the master and Fig. 5 shows the code snippet for the slave.

```
include <Wire.h>

struct UltrasonicSensor {
  int trigPin;
  int echoPin;
};

UltrasonicSensor sensors[4] = {
  {2, 3}, {4, 5}, {6, 7}, {8, 9}
};

const int detectionThreshold = 15; // cm
const int I2C_ADDRESS = 8;

unsigned long lastPrintTime = 0;
const unsigned long printInterval = 1000; // 1 second

delay(10); // prevent overload
}
```

Fig. 4: Code snippet for the master

```
#include <Wire.h>

// Traffic light pin setup
struct TrafficLight {
  int redPin;
  int greenPin;
};

TrafficLight lights[4] = {
  {2, 3}, // Lane 1
  {4, 5}, // Lane 2
  {6, 7}, // Lane 3
  {8, 9} // Lane 4
};

const int I2C_ADDRESS = 8;

// Time durations
const unsigned long GREEN_SHORT = 2000;
const unsigned long GREEN_LONG = 10000;
```

Fig. 5: Code snippet for the slave

Construction and testing

First, the master source code is uploaded to the Arduino Uno, and the slave source code to the Arduino Nano. After both programs are successfully uploaded, the circuit is assembled on the breadboard as shown in the wiring diagram (see Fig. 6). Fig. 3 and Table 3 should also be consulted during assembly for component placement.

All ultrasonic sensors should be carefully positioned to monitor traffic in one direction at the traffic junction. Before finalising the setup, all 5V and GND connections should be checked for continuity to avoid voltage drops.

During construction, the I2C connection between the two Arduinos must be made using pins A4 (SDA) and A5 (SCL), with both boards sharing a common ground for proper data exchange. Each ultrasonic sensor's Trig pin is connected to a digital output pin and the Echo pin is connected to a digital input pin on the Arduino Nano. The RGB LEDs representing the traffic lights are connected to the Arduino Uno via current-limiting resistors to prevent excessive current.

Wiring clarity is maintained by

TABLE 3

CONNECTIONS AMONG THE VARIOUS COMPONENTS

Arduino Uno	Arduino Nano	Sensor U4 Lane 1	Sensor U3 Lane 2	Sensor U3 Lane 3	Sensor U1 Lane 4	RGB	5V	GND
D2, D3		Trig, Echo						MOD1-GND
D4, D5			Trig, Echo					
D6, D7				Trig, Echo				
D8, D9					Trig, Echo			
	D2 D3					R - LED1 G - LED1	LED1-C	MOD2-G
	D4 D5					R - LED2 G - LED2	LED2-C	
	D6 D7					R - LED3 G - LED3	LED3-C	
	D8 D9					R - LED4 G - LED4	LED4-C	
SDA, SCL	A4, A5							
GND	GND	GND	GND	GND	GND			GND
5V	5V	5V	5V	5V	5V	5V	5V	